

“Empowering Science and Data Analytics through Python”

Lecture_02

by

Pranjal Saikia

MO ALIBAG

27 Apr to 01 May

Topic:

Conditionals and Loops:

- i) If**
- ii) elif**
- iii) else,**
- iv) For**
- v) while,**
- vi) break**
- vii) continue**

What is Conditionals in programming language?

Definition: Conditionals are statements that allow a program to make decisions based on conditions (true or false).

Example:

1. Real life example:

Ram stopped at traffic light. Now for Ram conditions are as follows:

- i) If the signal is red → stop
- ii) If the signal is green → go
- iii) If the signal is yellow → wait

How to put all these in a simple python program?

```
signal = "red"

if signal == "green":
    print("Go")
elif signal == "yellow":
    print("wait")
else:
    print("stop")
```

What is a loop in programming language?

Definition: A loop is a control structure in programming that is used to execute a block of code repeatedly as long as a given condition is satisfied or for a specified number of times.

Loops help in reducing repetition of code and make programs efficient.

Difference between Conditionals and Loops

Conditionals	Loops
Used for decision making	Used for repetition
Runs once	Runs multiple times
Uses if, elif, else	Uses for , while

Real Life Example:

Conditional: If it rains → take umbrella

Loop: Eat 3 biscuit → repeat action of eating three times.

Control statement: Control statements are statements in programming that control the flow of execution of a program, such as decision making, looping, and changing the normal sequence of execution.

Types of control statements:

1. Conditional (Decision making): if, elif, else
2. Looping (iteration): for, while
3. Control transfer: break, continue

if, elif, else, for, while, break, continue, together all these are called control statements because they control how a program executes.

Type	Keywords
conditionals	if, elif, else
loops	for, while
loop control	break, continue

1. if, elif, else

Application of conditionals

Practice:

Start from simple one:

1. Write a program to check if a number is positive.

```
number = - 5
if number >= 0:
    print(f "{number} is a positive number.")
else:
    print(f "{number} is not a positive number.")
```

2. Check if a number is even.

```
number = -5
if number % 2 == 0:
    print(f "{number} is a even number.")
else:
    print(f "{number} is not an even number.")
```

3. Find the largest of three numbers.

```
a = 2
b = 4
c = 1
if a >= b and a >= c:
    print(f "{a} is largest.")
elif b >= a and b >= c:
    print(f "{b} is largest.")
else:
    print(f '{c} is largest.')
```

Little advanced:

4. Write a code that take a user input: **marks**. If marks is greater than 75 then print output "Distinction", if marks is greater than 40 then print output "Pass". If less than 40 then print output "Fail".

```
marks = 75
if marks >= 75 :
    print(f "Distinction")
elif marks >= 40:
    1. print(f "Pass")
else:
    print(f"Fail")
```

5. Shyam is of age 25. He wants to play a computer game. The game has some initial inquiry asking the user to input the age. Use condition that
- i) if age is greater than equal to 20 print output will be “Shyam can play”.
 - ii) if age is less than 20 print output will be “Shyan cannot play”.
 - iii) if age is something else then “Enter a valid number”.

```
age = 25
if age >= 20:
    print(f "Shyam can play")
elif age < 20:
    print(f "Shyam cannot play")
else:
    print(f "Enter a valid number")
```

6. Take a use input an integer and print output “Greater” if the number is greater than 5. If the number is any other number including 5 then “smaller”

```
x = 25
if x >= 20:
    print(f "Greater")
elif age < = 5:
    print(f "Smaller")
```

2. for and while

Application of loops

Practice:

For loop:

1. Print numbers from 1 to 10.

```
for i in range(1,11):  
    print(i)
```

2. Print even numbers from 1 to 20.

```
for i in range(1,21):  
    if i % 2 == 0:  
        print(i)
```

3. Print sum of first 10 numbers.

```
total = 0  
for i in range(1,11):  
    total += i  
print('Sum =', total)
```

4. Count how many numbers are divisible by 3 from 1 to 30.

```
count = 0  
for i in range(1,31):  
    if i % 3 == 0:  
        count += 1  
print('Count =', count)
```

while loop:

1. Print numbers from 1 to 10.

```
i = 1
while i <= 10:
    print(i)
    i += 1
```

2. Print even numbers from 1 to 20.

```
i = 1
while i <= 20:
    if i % 2 == 0:
        print(i)
    i += 1
```

3. Print multiplication table of 5.

```
i = 1
while i <= 10:
    print('5 x', i, '=', 5*i)
    i += 1
```

4. Count digit in a number.

```
num = 56789
count = 0
while num > 0:
    count += 1
    num //= 10
print('Digits = ', count)
```

3. break and continue

Practice:

break

1. Stop printing when number = 5.

```
for i in range(1,11):  
    if i == 5:  
        break  
    print(i)
```

2. Find the first number divisible by 7.

```
for i in range(1,100):  
    if i % 7 == 0:  
        print('First divisible by 7: ', i)  
        break
```

3. Stop when negative number is entered.

```
While True:  
    num = int(input('Enter number: '))  
    if num < 0:  
        break  
    print('You entered:', num)
```

4. Find first even number in a list.

```
numbers = [3,7,9,12,15]  
for num in numbers:  
    if num % 2 == 0:  
        print('First even number:', num)  
        break
```


Continue

Practice:

1. Skip number 5 while printing 1 to 10.

```
for i in range(1,11):  
    if i == 5:  
        continue  
    print(i)
```

2. Print only odd numbers.

```
for i in range(1,21):  
    if i % 2 == 0:  
        continue  
    print(i)
```

3. Skip negative numbers in a list.

```
numbers = [5, -2, 8, -1, -10]  
for num in numbers:  
    if num < 0:  
        continue  
    print('Positive:', num)
```

4. Skip multiples of 3.

```
for i in range(1,16):  
    if i % 3 == 0:  
        continue  
    print('Positive:', num)
```

-----***-----